Introduction to ROAR High Performance Computing (HPC) System: Applications in Academic Research using R

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PennState

Background:
What is an HPC

High Performance Computing (HPC) System

- Supercomputer that Compute complex calculations and process large datasets
- ROAR = Penn State's Supercomputer
  - High performance research cloud
  - RISE – Research Innovations with Scientists and Engineers

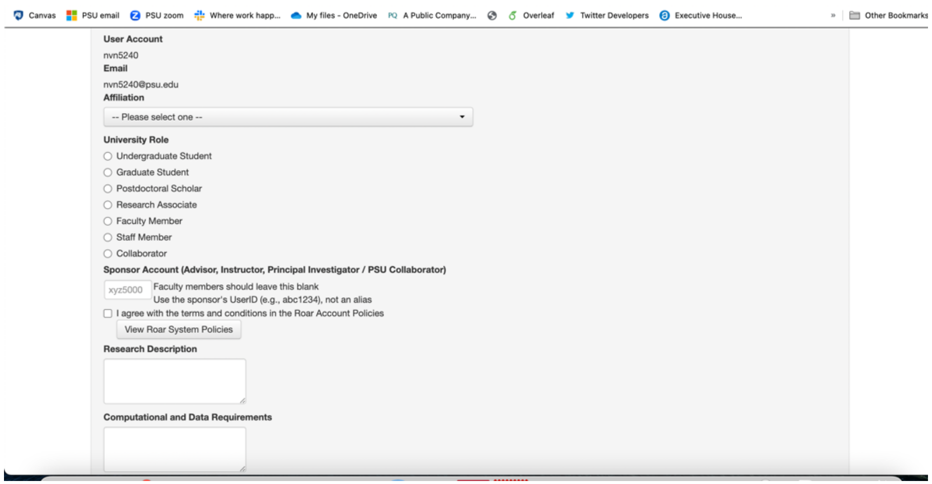
Goals

- Interface with HPC
- Submit a Job
  - Operating FileZilla
  - Writing a PBS script

Getting Started

Account Setup

- Request an Account:
  - https://www.icds.psu.edu/computing-services/account-setup/
- Submits a ticket to I-ASK
- All Non-faculty need a “sponsor” for a ROAR account
  - Advisor or Course Instructor

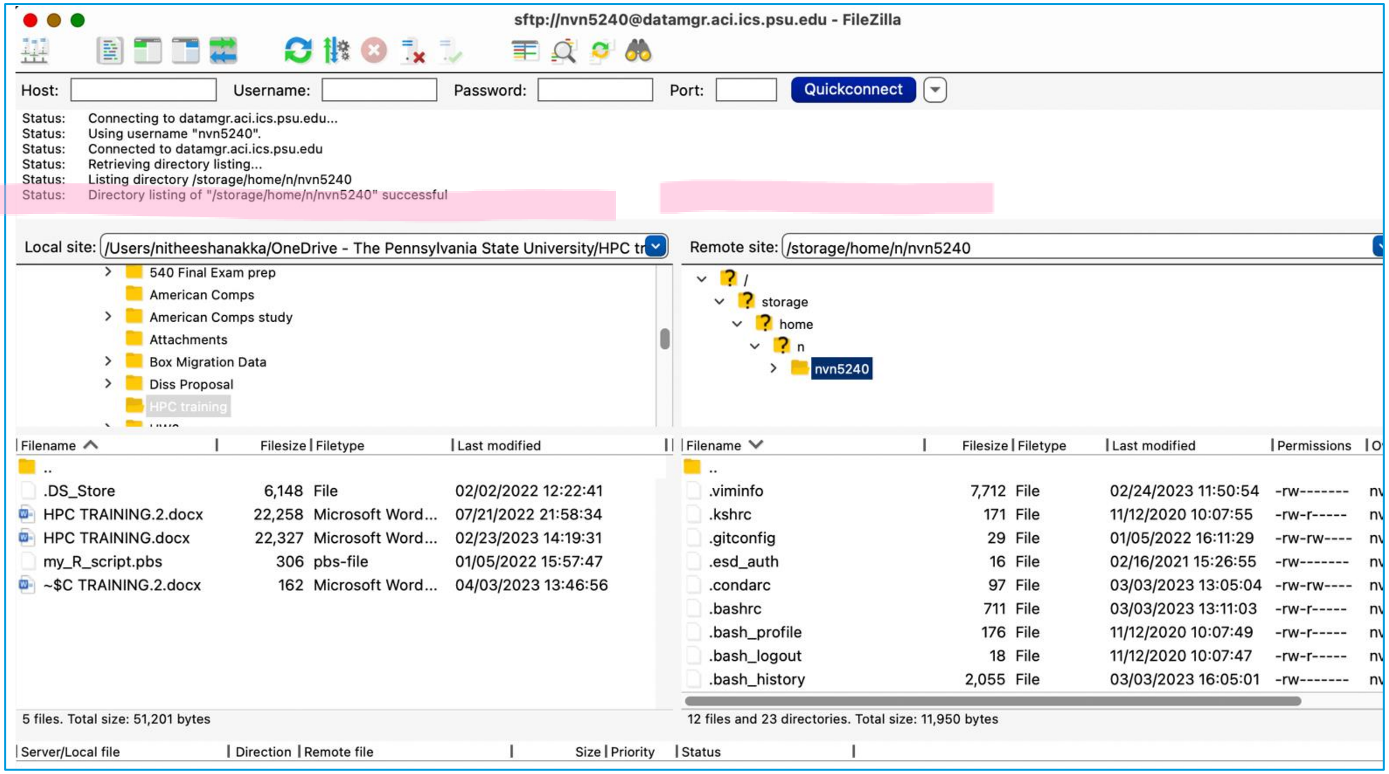


Resource: https://www.icds.psu.edu/computing-services/roar-user-guide/getting-an-account-penn-state-faculty-staff-and-students/

Getting Started

FileZilla *Client*

- Submit jobs and shift files *through* FileZilla
  - Operates like a gateway
  - Download here: https://filezilla-project.org/
- Other: command line, WinSCP Secure Protocol Client
  - Download here: https://winscp.net/eng/download.php



Submitting a Job
Step 0: Prepare Scripts

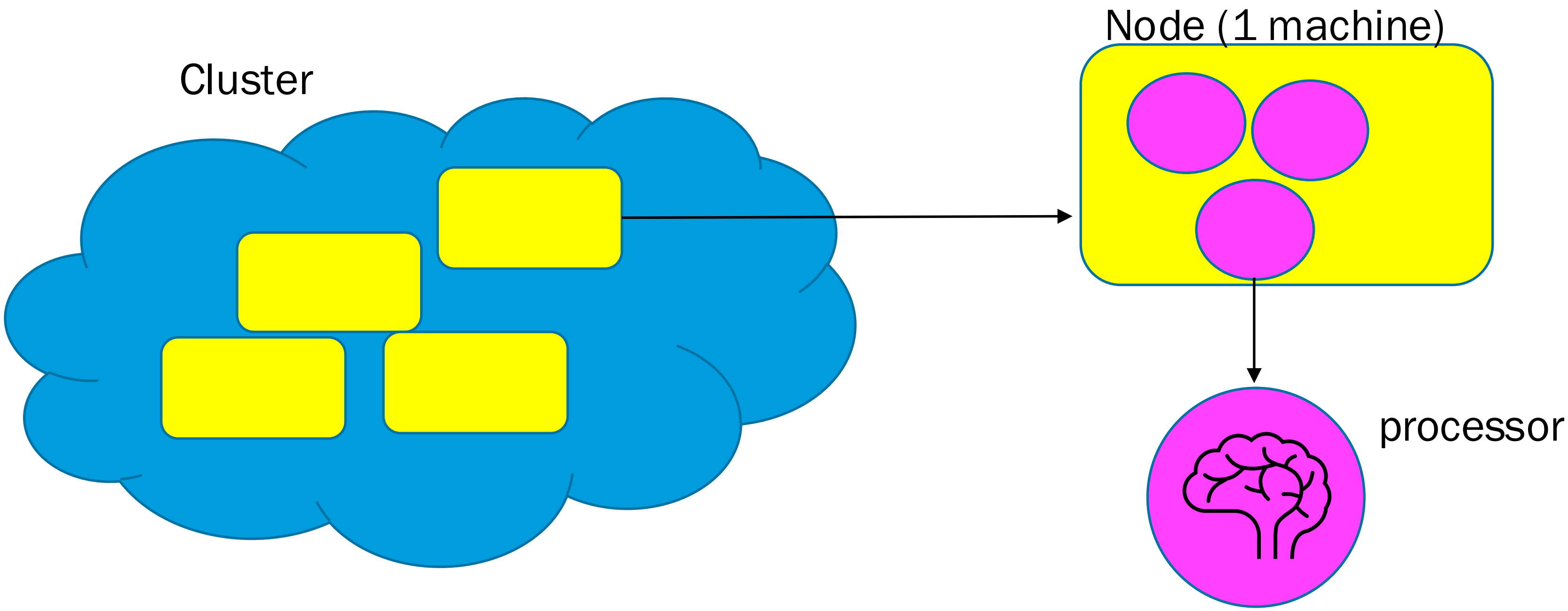
- Prepare R script
- Prepare PBS script: tells HPC what you need to complete job

```
#!/bin/bash -l
#PBS -N example
#PBS -l walltime=1:00:00
#PBS -l nodes=1:ppn=1
#PBS -l pmem=10gb
#PBS -j oe
#PBS -A open
#PBS -m abe nvn5240@psu.edu

# Go to the submission directory, making the folder that the pbs is in as the working dir
cd $PBS_O_WORKDIR

# Run the job itself
#R --file= /storage/home/...

R CMD BATCH example.R
```

Resource: https://www.icds.psu.edu/computing-services/roar-user-guide/how-to-assess-which-resources-you-should-request/

Submitting a Job
Step 0: Prepare Scripts

#!/bin/bash -l
#PBS -N example → What I want to call the job
#PBS -l walltime=1:00:00 → length of time requested for job (48 hours max in ”open”)
#PBS -l nodes=1:ppn=1
→ nodes = the number of computers
→ ppn = the number of processors per node



Submitting a Job
Step 0: Prepare Scripts

#!/bin/bash -l
#PBS -N example → What I want to call the job
#PBS -l walltime=1:00:00 → length of time requested for job
#PBS -l nodes=1:ppn=1
#PBS -l pmem=10gb → max memory per processor
#PBS -j oe → code output and error output should be in same file
#PBS -A open→ allocation (also known as “queue”)
#PBS -m abe nvn5240@psu.edu → emails you updates about job

Go to the submission directory, making the folder that the pbs is in as the working dir
cd $PBS_O_WORKDIR

Run the job itself
#R --file= /storage/home/…

R CMD BATCH example.R → run script and save output to a file

MAKE SURE R script and PBS script have SAME NAME!!

Submitting a Job
Step 1: Login

- Open Terminal
  - Mac- Search "terminal"
  - PC- search "command prompt"

- Mac: Type- ssh nvn5240@submit.aci.ics.psu.edu (save this)
  -> Type- Password
  -> Type- 1 – Duo Push Authentication
- Windows: PuTTy to access SSH Client
  - https://www.icds.psu.edu/computing-services/getting-started/
  - Install: https://www.putty.org/

- Arrive in **Home Directory**
  - Other storage directories that "live" here
    - work: quota limit = 128 GB
    - home: quota limit = 10GB
    - Scratch: 1 million file capacity but ITEMS DELETED IN 30 DAYS

Step 1a: Lets Look Around

Basic Bash Commands

- pwd = print working directory path
- ls = lists whats in a folder
  - ls -al= shows you when file is last modified
  - ls -a = shows all hidden files too
- cd [directory name] = change directory
  - Can also combine directory paths
  - cd .. = moving one directory "up"
  - cd ~ = move all the way back to home directory
- mkdir [new directory name] = make directory
- cat [file name] = look at script contents
- rm [file name] = remove a file , rm –r [directory name] = remove directory

- R = start R shell
  - q() = quits session
- python = start Python shell
  - exit() = quits session

Resource: https://www.earthdatascience.org/courses/intro-to-earth-data-science/open-reproducible-science/bash/bash-commands-to-manage-directories-files/

Submitting a Job
Step 2: Transfer Files through FileZilla

2a) First Login to FileZilla
- username= nvn5240
- password =
- Port = 22
- Host = datamgr.aci.ics.psu.edu
- After doing the above 1 time you can click "quick connect" to go directly to home directory
- 2-Factor authentication every login

2b) Drag & Drop
- Local → Remote
- Can also delete files/ folders in filezilla

- Note: Whatever changes you make either in local or remote DOES NOT automatically make those changes on the "opposite" computer

Submitting a Job
Step 3: Submit a Job

PBS File submits the job!

- Be in the folder with the script and PBS files are in
- qsub [NAME OF PBS FILE]
  - qsub [Allocation Name] [Name of pbs file]

Step 3a : Check job status

- qstat -u nvn5240
  - q = job is waiting in line, R = running, C = complete

- Step 3b : delete a job
  - qdel [job number]

Installing R Packages

A) Put it in your R script

B) Install in R session in terminal

C) Conda Installer – eg. ”Tidyverse”

- 1) module use /gpfs/group/RISE/sw7/modules- this is where anaconda lives
- 2) module load anaconda
- 3) conda create -n rpkgs *(Will not have to do this again for future installs)*
  - this name is in some default location
- 4) conda activate rpkgs
- 5) Conda install -c conda-forge r-tidyverse
  - https://anaconda.org/conda-forge/r-tidyverse

Resources

Programming Help:
https://www.icds.psu.edu/computing-services/roar-training-resources/

ROAR User Guide
https://www.icds.psu.edu/computing-services/roar-user-guide/
ICDS Guidelines
https://www.icds.psu.edu/wp-content/uploads/2020/04/ICDS-ACI-P030-Access-Control-Formatted_v3.1.pdf

Parallelization:
doParallel R package:
https://cran.r-project.org/web/packages/doParallel/vignettes/gettingstartedParallel.pdf
foreach():
https://cran.r-project.org/web/packages/foreach/vignettes/foreach.html

I-Ask Help Desk Virtual Office hours

Myself
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Github: https://github.com/NitheeshaN